

The SPO portfolio tree: model v1

The model and its training, with the equations; then what the planted-threshold experiment says

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Outline

1. What we do, and how we test it
2. The model: the tree, the decision problem, the loss, the training
3. Model v1: the split-search options and the frozen configuration
4. The planted-threshold experiment: recovery, performance, trading rate, the options, v1 on 100 datasets, what can be found, v1 against signal, volatility and half-life
5. A fair assessment, and what comes next

What we do, and how we test it

Problem

- weights of n assets every h days, from features x_t , with fees
- a decision problem, not a forecast

Model

- small tree $\rightarrow$ leaf scores $\rightarrow$ mean–variance optimizer with fees
- loss: decision regret through the optimizer
- splits confirmed on the tree's own traded path
- Out of sample Sharpe pruning

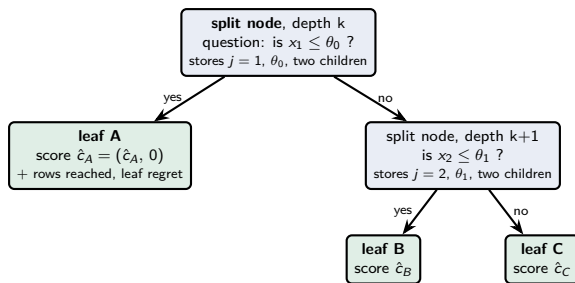
Test

- synthetic market, one feature x , planted threshold d
- benchmark: *ideal rule* = optimizer fed $\mathbb{E}[r \mid x]$
- constant volatility; asset vs cash; weekly unless stated

Model v1

- frozen 24 September 2026
- $\approx 15,000$ fitted trees behind the numbers
- validated on 100 datasets

The tree: what a node stores



- tree = nested questions “is $x_j \leq \theta$?”, ending in leaves
- **split node**: feature j , threshold θ , two children; no score
- **leaf**: one score vector $\hat{c}_\ell = \left(\hat{c}_\ell^{(1)}, 0 \right)$ (only the difference matters); rows reached; leaf regret
- holdings u_t , weights w_t , returns: per date, not per node
- depth $\leq 3 \Rightarrow \leq 8$ leaves $\Rightarrow \leq 8$ scores
- growth: leaf $\rightarrow$ split node + 2 leaves; pruning: the reverse
- use: $x_t \rightarrow$ leaf $\rightarrow \hat{c}_\ell \rightarrow$ optimizer $\rightarrow w_t$

The decision: mean–variance with proportional fees

Notation

- u : holdings before the decision,

$$u_{t+1} = \frac{w_t \circ (1 + r_t)}{1 + r_t^\top w_t}, \quad u_1 = \mathbf{1}/n$$
- c : the tree's scores, in units of the h -day return; cash = 0 (only the score differences matter)
- Σ : covariance of the h -day return = $h \times$ the covariance of the last 180 daily returns; $\sigma_h^2 = h\hat{\sigma}^2$
- f : fee per traded notional, 3 bp = 0.03% on the asset, 0 on cash; $\lambda = 1$: risk aversion, weight of the h -day variance
- $\mathcal{W} = \{w : 0 \leq w_i \leq 1, \sum_i w_i = 1\}$

$$w^*(c; u, \Sigma) = \arg \max_{w \in \mathcal{W}} \underbrace{c^\top w}_{\text{gain}} - \underbrace{f^\top |w - u|}_{\text{fees}} - \underbrace{\lambda w^\top \Sigma w}_{\text{risk}}$$

Asset vs cash (w = asset weight, $\kappa = 2\lambda\sigma_h^2$):

$$w^* = \begin{cases} \min\{(c - f)/\kappa, 1\} & c - \kappa u > f \text{ (buy)} \\ \max\{(c + f)/\kappa, 0\} & c - \kappa u < -f \text{ (sell)} \\ u & |c - \kappa u| \leq f \text{ (no trade)} \end{cases}$$

v1 (5-day returns): $\kappa = 2 \cdot 1 \cdot 5 \cdot (1\%)^2 = 0.1\%$; $w : 0 \rightarrow 1$ for $c : 0.03\% \rightarrow 0.13\%$; leaf scores $\pm 0.3\text{--}0.5\% \Rightarrow w \in \{0, 1\}$; continuous sizing needs $\lambda > 5$

Properties

- strictly concave $\Rightarrow$ unique w^*
- $w^*(c + \alpha \mathbf{1}) = w^*(c)$: only score differences matter
- exact solution: 5 states per weight (0, cap = $\bar{w}$, u_i , buy, sell) $\Rightarrow 5^n$ linear systems, best feasible one

The loss: decision regret; the score of a leaf

$$U_t(w) = r_t^\top w - f^\top |w - u_t| - \lambda w^\top \Sigma_t w$$

$$\text{Reg}_t(c) = U_t(w^*(r_t; u_t, \Sigma_t)) - U_t(w^*(c; u_t, \Sigma_t)) \geq 0$$

hindsight decision – decision on c (given the same holdings u_t)

$$\hat{c}_\ell = \arg \min_{c \in [-b, b]^n} \sum_{t \in \ell} \text{Reg}_t(c), \quad b = 10\% \text{ per holding period, bound} + \text{resolution}$$

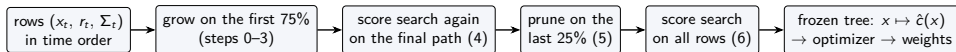
Leaf score

- one regret per date, summed over the leaf's dates
- bounded coordinate search: 3 passes $\times$ 7 values
- not a forecast: same regret for any score on the right side

Consequences

- fees $\Rightarrow$ regret depends on $u_t \Rightarrow$ holdings from a replay of the tree
- only differences: relative returns, asset vs cash $\Rightarrow$ help to choose the right features
- same Σ_t as the optimizer: unseen risk not learnt

Training: replay, two growth gates, out of sample pruning



0. root score from $u_t = 1/n$; **replay** (trade the scores date by date from u_1) $\Rightarrow u_t$, Sharpe S ; holdings frozen for the round
1. **regret gate**, per leaf: score search again from current u_t ; candidates; children of the 3 best; keep iff $\Delta/N_\ell > 10^{-6}$, $\Delta = L_\ell - L_{\text{left}} - L_{\text{right}}$ (regret per date, in h -day return)
2. **Sharpe gate**: rank all kept splits by Δ ; whole-tree replay of the first; accept iff $S' > S$, else next; one per round; none $\Rightarrow$ stop
3. replay the new tree $\Rightarrow$ new u_t ; back to 1 (depth ≤ 3 ; ≥ 20 rows and $\geq 10\%$ of the parent per leaf)
4. score search again on every leaf, final path; kept iff S does not fall
5. **pruning**, bottom-up, on the last 25% (never used for growth): with vs without the split, both replayed; $g_t = \text{Reg}_t(\text{without}) - \text{Reg}_t(\text{with})$; keep iff $\bar{g} > 10^{-6}$, $\bar{g} > k_p \text{se}(\bar{g})$ [$k_p = 2$ below the root, 0 at the root], checking Sharpe $\uparrow$
6. score search again on all training rows

Gates 1–2: in-sample. Gate 5: out-of-sample.

Split-search options: quantile candidates, refinement, soft splits

Common: admissible candidates $\theta_{(1)} < \dots < \theta_{(K)}$ (each side $\geq m_* = \max(m, \lceil \varphi N \rceil)$ rows; $m = 20$, $\varphi = 10\%$); cheap screen

$$Q(\theta) = \sum_{t \in I} \text{Reg}_t(c_{\text{side}(t)}(\theta)), \quad c_{\text{left}}(\theta) = \text{clip}(\text{mean}\{r_t : x_t \leq \theta\}, \pm b), \text{ idem right}$$

Deciles by rank + refinement

- keep ranks $\text{round}(q(K-1))$, $q = 0.1, \dots, 0.9$: 9 screens instead of $K \approx N - 2m_*$
- refinement: screen every candidate between the two deciles around the best one, θ_o ; best of them $\theta_\bullet$
- replace iff $Q(\theta_o) - Q(\theta_\bullet) > k \text{ se}$, $\text{se} = \sqrt{|I|} \text{std}_t(\delta_t)$, $k = 1$
- 3 best survivors $\rightarrow$ full fit of the children

Soft split (off in v1: $\kappa = 0$)

$$z_k = \frac{Q(\theta_k) - \min_I Q(\theta_I)}{\text{se}_k}, \quad \pi_k = \frac{e^{-z_k/\kappa}}{\sum_I e^{-z_I/\kappa}}$$

- all candidates kept, weighted; Gaussian variant $e^{-z^2/2\kappa^2}$
- membership: left with $\mu(x) = \sum_{k: \theta_k \geq x} \pi_k$, right with $1 - \mu(x)$; weights $\omega_{t\ell}$ multiply down the tree
- leaf fit: $\min_c \sum_t \omega_{t\ell} \text{Reg}_t(c)$; row score $\hat{c}(x_t) = \sum_\ell \omega_{t\ell} \hat{c}_\ell$: a ramp, not a jump
- $\kappa \rightarrow 0$: plain split

Model v1: the frozen configuration (24 September 2026)

Choice	v1	Why (section of the experiments report)
risk aversion λ	1 (weight of the h -day variance)	positions 0/1: regime classifier, not sizer (8)
fee f	3 bp = 0.03% of the traded notional on the asset; 0 on cash	market-sized; 10 bp starves fast rates (8)
decision rate h	weekly ($h = 5$ trading days); every 2 days as alternative	daily $\approx$ weekly; monthly loses (4)
depth	≤ 3	depth alone changes nothing (8)
leaf minimum	20 decisions and 10% of the node	tail leaves at 0.7% of the data (7)
candidate thresholds	deciles by rank + refinement (1 s.e.)	= base Sharpe, steadier thresholds (9, 12)
soft splits	off	never raise the Sharpe at $\lambda = 1$ (9, 12)
growth gates	regret reduction, then replayed Sharpe $\uparrow$	kept as designed; v2 revisits
pruning	last 25%; hurdle 2 s.e. below the root, none at the root	out-of-sample judge (7); removes spurious leaves at no cost (12)
covariance	$\Sigma_t = h\hat{\sigma}_t^2, \hat{\sigma}_t^2$ the variance of the last 180 daily returns	20-day window: noisier than timely (regime report)
conventions	cash column; all-cash Sharpe = 0; $ \hat{c} \leq 10\%$ per holding period	

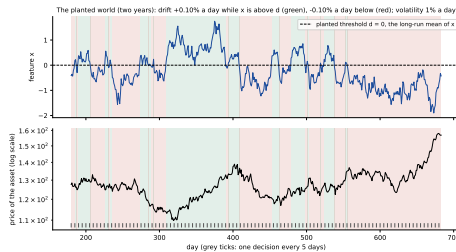
Base of every experiment = same pipeline, every option off: depth 2, all candidates screened, plain pruning.

The test-bed: a market with a planted answer (constant volatility)

$$dX_t = k(\mu - x_t)dt + \sigma dW_t, \quad dW_t \sim \mathcal{N}(0, dt)$$

$$\frac{dS_t}{S_t} = \mu(X_t)dt + \sigma dW_t, \quad \mu(X_t) = \pm e\sigma \text{ as } x_t \gtrless d$$

σ (daily)	1% (16% a year), both regimes
$e = \mu/\sigma$ (daily)	0.10, 0.05, 0.02: drift ± 25 , ± 13 , $\pm 5\%$ a year; Sharpe of the regime $e\sqrt{252} = 1.6, 0.8, 0.3$
k (daily)	0.02: half-life $\ln 2/k = 35$ days; 0.01 to 0.10 (69 to 7 days)
m, s^2, d	0, 1, 0: x standardized (mean 0, sd 1), threshold at its mean
h (trading days)	1, 2, 5, 10, 21: daily to monthly
data	daily steps (t in trading days); 15 years of training (3,780 days), 5 of test (1,260)

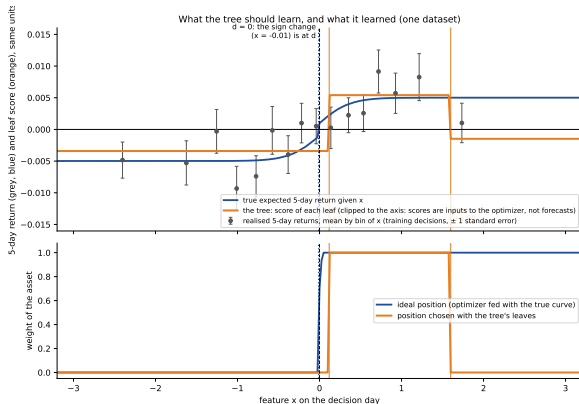


Measured on 40–100 datasets per cell (Monte Carlo)

- first threshold vs planted $d = 0$ (x has sd 1)
- annualized test Sharpe: tree vs ideal rule ($\mathbb{E}[r | x]$ fed to the optimizer), rule at d , always the asset
- share of the ideal gain captured

Second market (volatility follows a threshold too): own report; one line at the end.

One fitted tree (weekly, edge 0.10)



What is drawn (x on the decision day)

- **blue**: the true expected 5-day return given x ; grey: the realised returns, averaged by bin of x
- **orange**: the tree's score, one value per leaf, in the same units
- **bottom**: the position each one implies

Key points

- **The root split lands on the planted threshold.**
- **Scores are not forecasts.** They sit far outside the optimizer's narrow window, so the positions match the ideal rule wherever their sign is right.
- **The far-right leaf is spurious.** A tail of x sent to cash while the ideal rule stays invested; v1's pruning hurdle removes such leaves.
- **At $\lambda = 1$ the ideal rule is a step too.** The threshold is the whole problem.

Result 1: the threshold is recovered at market sizes (E2, base pipeline)

edge	h	split	threshold	annual. test Sharpe		
		on x (%)	within 0.25	tree	ideal	gain
0.10	1	97	95%	0.80	1.03	76%
	5	94	81%	0.73	0.88	79%
	21	81	60%	0.54	0.68	75%
0.05	1	82	60%	0.12	0.49	25%
	5	78	53%	0.24	0.43	51%
	21	62	34%	0.17	0.35	43%
0.02	5	54	26%	-0.03	0.15	-3%

- row = 100 datasets (50 at edge 0.02), threshold planted at 0; edge e = daily drift / daily volatility: 0.10, 0.05, 0.02 = $\pm 25\%$, $\pm 13\%$, $\pm 5\%$ a year; h in trading days
- split on x : trees keeping a split (else: no split at all)
- within 0.25: first threshold within 0.25 sd of x from 0
- Sharpe: mean over datasets, s.e. ≈ 0.04
- gain: tree's excess test return over the no-split tree, as a share of the ideal rule's

- $\pm 25\%/yr$: **found at every rate**; 95% within 0.25 daily, 60% monthly; 75–79% of the ideal gain
- $\pm 13\%/yr$: **graceful degradation**; 62–82% split; 25–51% of a small gain (ideal 0.35–0.49)
- $\pm 5\%/yr$: **nothing to find in 15 years**; half no split, half at random; ideal rule ≈ 0.15 ; finding nothing = right answer
- daily $\approx$ weekly; monthly loses (regime changes inside the holding period)

Result 2: the strategy's performance out of sample (E2, base pipeline)

edge	h	annualized test Sharpe ratio				gain
		tree	ideal rule	rule at d	buy & hold	captured
0.10	1	0.80	1.03	1.04	-0.02	76%
	2	0.86	0.97	0.98	-0.02	85%
	5	0.73	0.88	0.88	-0.02	79%
	10	0.63	0.78	0.78	-0.02	78%
	21	0.54	0.68	0.66	-0.02	75%
0.05	1	0.12	0.49	0.49	-0.02	25%
	2	0.21	0.46	0.46	-0.02	44%
	5	0.24	0.43	0.42	-0.02	51%
	10	0.17	0.39	0.38	-0.02	43%
	21	0.17	0.35	0.34	-0.02	43%

- 100 datasets per row; s.e. ≈ 0.04
- ideal rule: optimizer fed $\mathbb{E}[r \mid x]$; rule at d : asset iff $x \geq d$;
buy & hold: always fully in the asset, never in cash
- turnover per weekly decision (edge 0.10, both legs): tree 0.25,
ideal 0.29 $\Rightarrow$ fees $\approx 0.2\%$ a year (3 bp on the asset leg)

- **The tree captures most of what the signal offers.** With a $\pm 25\%$ regime it earns about three quarters of the ideal rule's gain.
- **Learning the threshold is the whole job.** A rule that only knows the planted threshold matches the ideal rule: at $\lambda = 1$ there is nothing else to learn.
- **What is lost is precision.** The gap to the ideal rule is the threshold error plus a few spurious leaves.
- **A weak regime is a limit of the data, not of the tree.** At $\pm 13\%$ a year the tree keeps only part of a small gain: 15 years barely identify it.
- **Without the signal there is nothing to earn.** The asset drifts up in one regime and down in the other, so buying and holding earns zero: all the gain comes from switching with x .

Trading rate against the speed of the feature (edge 0.10, 50 datasets per cell)

half-life of x	daily		weekly		monthly	
	tree	ideal	tree	ideal	tree	ideal
69 days ($k = 0.01$)	0.63	1.09	0.84	0.97	0.68	0.84
35 days ($k = 0.02$)	0.80	1.03	0.73	0.88	0.54	0.68
14 days ($k = 0.05$)	0.69	0.98	0.65	0.75	0.30	0.49
7 days ($k = 0.10$)	0.77	0.96	0.45	0.60	0.18	0.34

annualized test Sharpe ratios, tree and ideal rule

- daily $\approx$ weekly: more decisions = more fees, not more information
- monthly loses, more so for a fast feature: half-life 7 d $\Rightarrow$ 0.18 vs 0.34
- slow feature + daily decisions: over-trading (0.63 vs 1.09)
- $\Rightarrow$ weekly in v1

What the design options bought (E3–E5, E8; 40 datasets per cell, edge 0.10)

setting	annual. Sharpe, tree	
	weekly	2-day
base (every option off)	0.80	0.87
depth 3	0.80	0.89
$\lambda = 5$ / $\lambda = 10$	0.70 / 0.65	0.81 / 0.71
fee 10 bp	0.60	0.53
deciles + refinement	0.82	0.82
soft splits $\kappa = 1$	0.53	0.35
soft splits $\kappa = 0.25$	0.73	0.61
hurdle 2 s.e., every node	0.43	0.47
hurdle 2 s.e., below the root only	0.77	0.74
deciles + refin. + hurdle below root	0.80	0.79

base = the tree and training of Results 1–2 with every option off: depth ≤ 2 , every threshold screened, pruning without a hurdle, $\lambda = 1$, fee 3 bp. Each other row changes one thing from it. Ideal rule 0.91 weekly, 1.00 at 2 days; s.e. ≈ 0.04 ; last row = v1 at depth 2

- **no option beats the base beyond noise**; what is left is threshold noise
- depth 3: nothing (a maximum); $\lambda = 5$ –10: gradual positions, learnt; 10 bp: fast rates stop trading
- deciles + refinement: = base, 9 candidates instead of hundreds, steadier thresholds $\Rightarrow$ **in v1**
- soft splits: $\kappa = 1$ score in the no-trade band; $\kappa = 0.25$ = base from weekly on, loses at 2 days $\Rightarrow$ **off** (smoothing the score $\neq$ sizing the position at $\lambda = 1$)
- hurdle at every node: loses true root splits (25% block has no power at 2 s.e.) $\Rightarrow$ **off**
- hurdle below the root: second-level leaves 25/10/5/5% $\rightarrow$ 5/0/0/0%, no cost from weekly on $\Rightarrow$ **in v1**

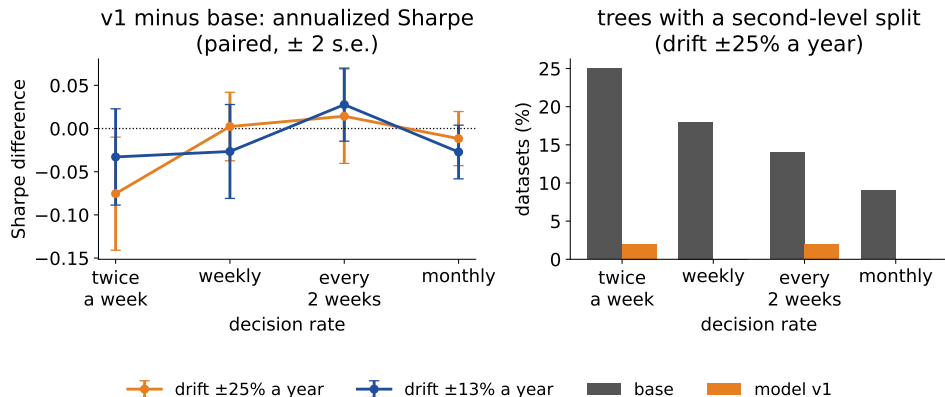
Model v1 against the base, on the same 100 datasets (E9, 800 fits)

edge	h	leaves		2nd level	within 0.25	Sharpe		v1 — base (2 s.e.)
		base	v1	base → v1	base → v1	base	v1	
0.10	2	2.24	2.00	25 → 2%	92 → 89%	0.86	0.78	−0.08 (0.07)
	5	2.14	1.94	18 → 0%	81 → 82%	0.73	0.73	+0.00 (0.04)
	10	2.03	1.92	14 → 2%	70 → 72%	0.63	0.64	+0.01 (0.05)
	21	1.90	1.81	9 → 0%	60 → 64%	0.54	0.53	−0.01 (0.03)
0.05	2	2.18	1.72	38 → 4%	64 → 57%	0.21	0.18	−0.03 (0.06)
	5	2.01	1.80	21 → 3%	53 → 51%	0.24	0.22	−0.03 (0.05)
	10	1.93	1.69	19 → 1%	47 → 49%	0.17	0.20	+0.03 (0.04)
	21	1.76	1.59	13 → 0%	34 → 42%	0.17	0.14	−0.03 (0.03)

same 100 datasets (seeds 0–99); Sharpe: annualized test Sharpe ratio;
 2nd level: datasets keeping a second-level split; last column: mean paired
 difference of annualized test Sharpe, 2 s.e. in brackets; half of the
 datasets: identical Sharpe

- **The question.** Does the frozen v1 do better than the base it was built from? Both are fitted on the same 100 histories, so every difference is paired.
- **Same Sharpe.** From weekly decisions on, v1 and the base are within noise of each other.
- **Cleaner trees.** The spurious second-level leaves are gone, the threshold is found as often, and the trees are smaller.
- **One cost.** Deciding every 2 days, v1 loses a little: some second-level splits are real there, and the pruning hurdle removes them.
- **Verdict: v1 is the base with a cleaner structure, not a higher Sharpe.**

Model v1 against the base: the paired differences in one look (E9)



Same 100 datasets for v1 and the base (seeds 0–99), 15 years of training, 5 of test. Left: annualized test Sharpe ratio of v1 minus that of the base, dataset by dataset; mean and two standard errors. Right: share of the datasets whose tree keeps a second-level split, at a drift of $\pm 25\%$ a year.

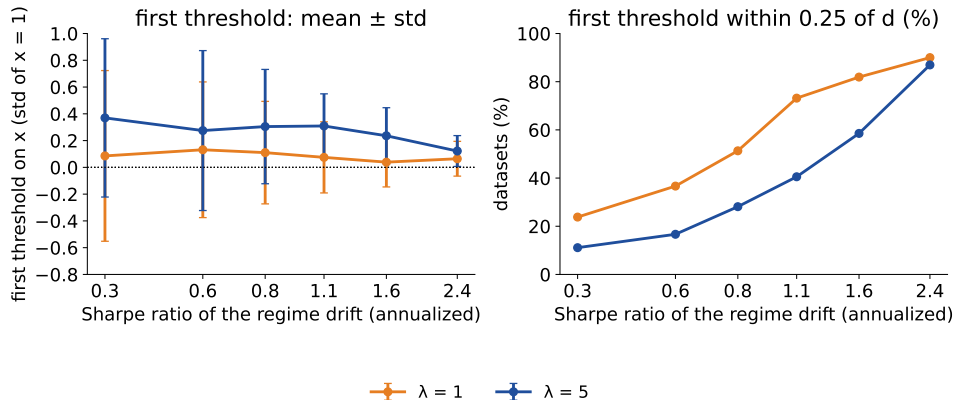
What can be found at all: resolution against volatility and feature noise (E6)

sweep (weekly)	value	edge	median $ \theta $	tree / ideal
σ per day (a year)	0.4% (6%)	0.25	0.05	1.85 / 2.10
	1% (16%)	0.10	0.12	0.69 / 0.87
	2% (32%)	0.05	0.39	0.23 / 0.42
	4% (63%)	0.025	0.95	-0.05 / 0.18
$x + \tau \varepsilon$ observed	0	0.10	0.12	0.69 / 0.87
	0.25	0.10	0.19	0.66 / 0.79
	0.5	0.10	0.25	0.45 / 0.66
	1	0.10	0.55	0.26 / 0.48
	2	0.10	1.15	0.09 / 0.29

60 datasets per row; volatility sweep at a fixed drift of $\pm 25\%$ /yr;
 $|\theta|$: first threshold's distance from the planted 0 (sd of x);
 $\varepsilon \sim \mathcal{N}(0, 1)$ daily white noise, τ its std in std of x ($= 1$); the
price follows the true x ; tree / ideal: annualized test Sharpe

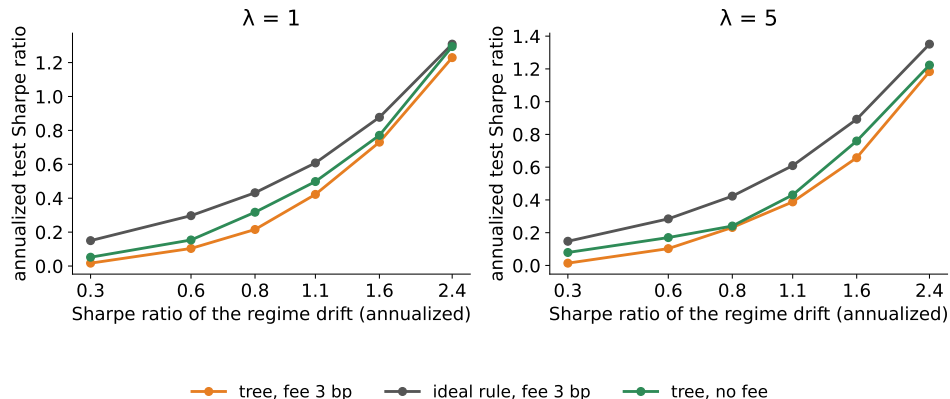
- threshold error $\approx 0.012/e$ down to $e \approx 0.05$; below: a guess
- noise τ on $x \approx$ edge divided by $1 + \tau^2$
- $\tau \leq 0.25$: free; 0.5: error doubles; 1: ideal Sharpe halves
- edge below a few bp/day: unlearnable from 15 years, by any method
- $\Rightarrow$ pre-test every real feature with this protocol

E10: the planted threshold against the strength of the signal



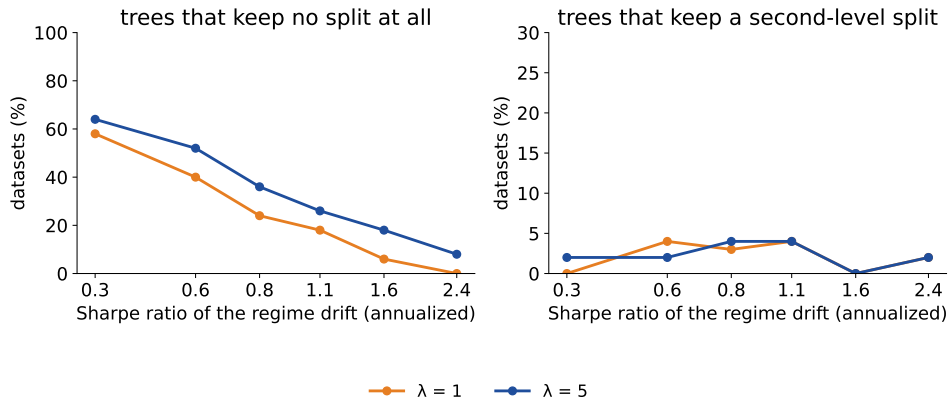
Model v1, weekly decisions, fee 3 bp, 16% volatility, half-life of x 35 days. The drift goes from ± 5 to $\pm 38\%$ a year, that is a Sharpe ratio of the regime drift from 0.3 to 2.4; 50 datasets per point (100 at 0.8 and 1.6). Left: the first threshold found, mean and one standard deviation over the datasets that split; the planted value is 0. Right: share of the datasets whose first threshold is within 0.25 of it.

E10: the Sharpe ratio against the strength of the signal, with and without fees



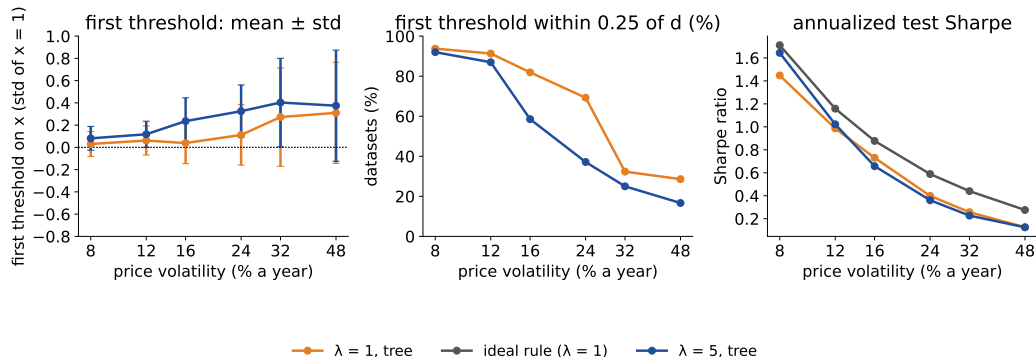
Same sweep. Annualized test Sharpe ratio of the tree with the 3 bp fee, of the tree without fees, and of the ideal rule (the optimizer fed with the true expected return), at $\lambda = 1$ and $\lambda = 5$.

E10: trees that keep no split, trees that keep a second-level split



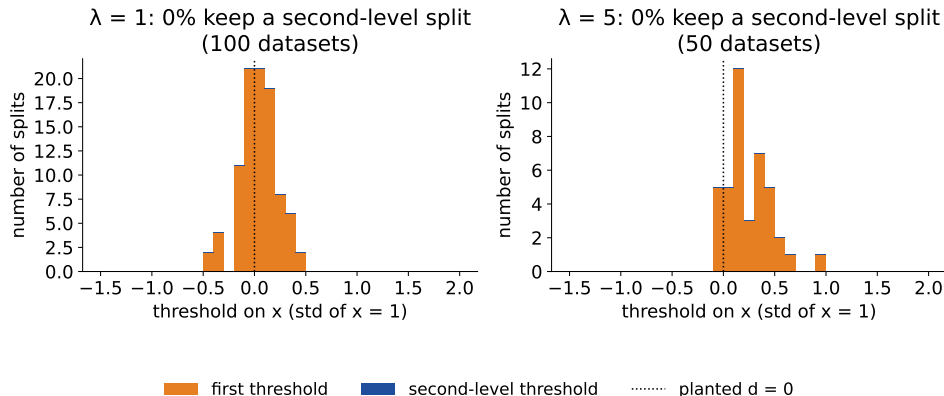
Same sweep, fee 3 bp. Left: share of the datasets where pruning leaves a single leaf. Right: share where a split below the root survives; the market has one threshold, so such splits are spurious.

E10: the price volatility, at a fixed drift of $\pm 25\%$ a year



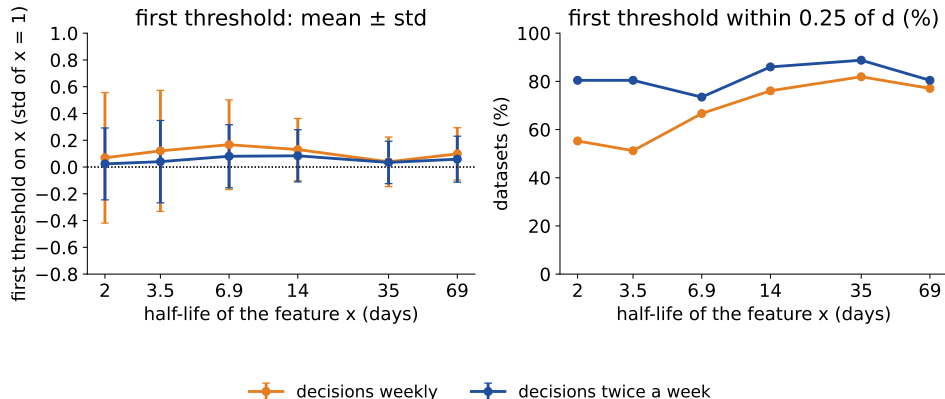
Model v1, weekly, fee 3 bp, half-life 35 days, 50 datasets per point. The drift stays at $\pm 25\%$ a year while the volatility goes from 8 to 48% a year, so the Sharpe ratio of the regime drift falls from 3.2 to 0.5. Grey: the ideal rule at $\lambda = 1$.

E10: where the second-level thresholds land



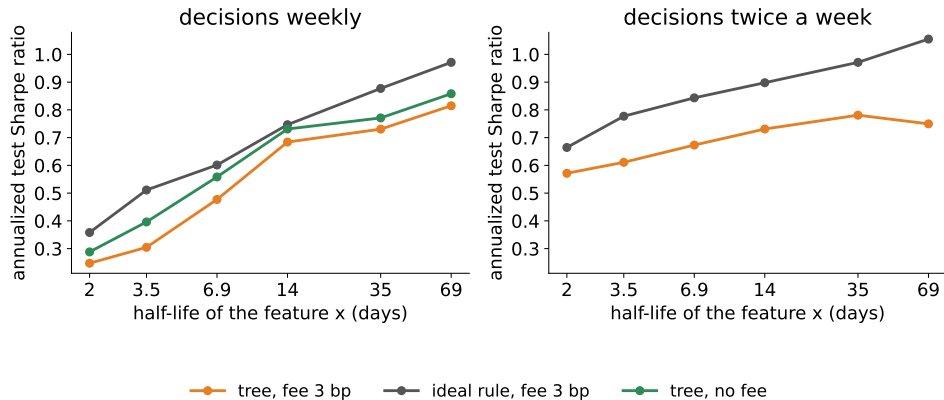
Model v1, weekly, drift $\pm 25\%$ a year, 16% volatility, fee 3 bp. Every threshold the trees keep: the first thresholds in orange, the second-level ones in blue. There are none: v1's pruning hurdle removes every split below the root in this market.

E10: the half-life of the feature: the threshold



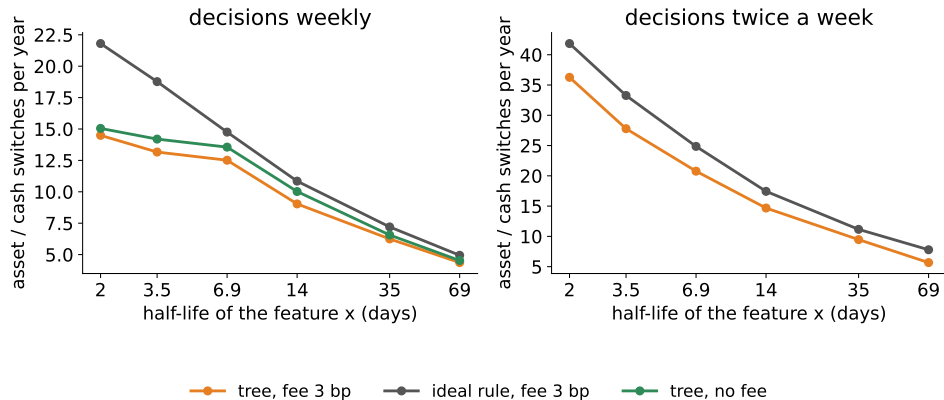
Model v1, $\lambda = 1$, fee 3 bp, drift $\pm 25\%$ a year, 16% volatility, 50 datasets per point. The half-life of x goes from 2 to 69 days (35 days everywhere else in this deck); decisions weekly or twice a week.

E10: the half-life of the feature: the Sharpe ratio



Same sweep. Annualized test Sharpe ratio of the tree and of the ideal rule; at weekly decisions also the tree without fees.

E10: the half-life of the feature: how often the tree switches



Same sweep. Number of full moves between the asset and cash per year on the test period, tree and ideal rule. At 3 bp a full move costs 0.03%, so 36 a year cost about 0.1% a year.

E10 in five points

- **The Sharpe ratio of the regime drift sets the precision.** A $\pm 25\%$ regime is found within 0.25 of the threshold in 8 datasets out of 10, a $\pm 5\%$ one in 1 out of 4, and then half the trees keep no split. Fees change almost nothing.
- **Volatility acts through that Sharpe ratio**, until 32–48% a year, where the risk term makes the ideal position gradual and the recovery drops below what the Sharpe ratio alone predicts.
- **At $\lambda = 5$ the split sits above the threshold, on purpose.** The ideal position is a ramp in x ; the best single split cuts it in the middle, not at d .
- **Second splits are gone.** At most 8% of the datasets keep one.
- **A fast feature needs a fast decision rate, and fees are not the limit.** With a half-life of a few days, weekly decisions cannot follow the regime; the tree switches less often than the ideal rule, not more. Deciding twice a week recovers the threshold and most of the ideal Sharpe at about 0.1% a year of fees. A slow feature is best traded weekly.

A fair assessment, and what comes next

Holds

- $\pm 25\%$ regime, monthly feature: recovered at every rate; 75–85% of the attainable gain
- $\pm 13\%$: graceful; $\pm 5\%$: correctly nothing
- v1 = base with smaller trees, steadier thresholds (100 paired datasets); cost 0.08 at 2 days
- second market (volatility regime): drift threshold found better when the calm regime pays; a variance-only boundary not learnt (lagged covariance in the regret)

Weak or untested

- no option beats the base beyond noise: structure, not Sharpe
- growth Sharpe gate: in-sample, mixes a risk preference into the structure
- one clean feature; synthetic data only; decoys only at an unrealistic edge; depth budget unused

Next (v2)

- growth gate off / replayed regret instead; same 100 datasets
- regime-aware variance for optimizer and regret
- two features + decoys at market-sized edges
- real data; this protocol as the pre-test of every feature